

# Supervised Bayesian Matrix Factorization Identifies Prognostic Gene Expression Programs

## Project Update

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June 18, 2026

## Agenda

**Background** · The PDAC prognostic problem · why supervise the factorization

**Method updates** · The joint model · two linear predictors ( $L$  vs  $(YF)$ ) · explicit priors & non-parametric baseline hazard · the ELBO & variational family · CAVI updates & convergence · preprocessing · selecting  $K$  · the cohort-membership extension

**Results** · Multi-cohort simulation study · real-data gene programs & survival · external validation on five held-out cohorts

## Impact & Future Directions

Roadmap for the next ~25 minutes. The methods section directly works through the feedback from April: explicit priors, the non-parametric baseline hazard, defining the variational family, the convergence criterion, the preprocessing fix for  $p$  n, no-leakage normalization, CV-based  $K$  selection, and the risk-direction correction.

## Background

### Background: PDAC needs molecular prognosis

- **Five-year survival**  $< 12\%$  — among the most lethal solid tumors
- Most patients diagnosed late; **no widely adopted molecular stratification** at diagnosis
- Bulk expression mixes malignant, stromal, and immune signals — prognostic structure is interwoven and often **low-variance**
- **Goal:** recover *gene expression programs (GEPs)* whose activation predicts survival, and validate them on **independent** cohorts
- A GEP is a coordinated set of genes (a column of  $F$ ) whose patient-level activation (a column of  $L$ ) we test for survival association

**i** Why model expression and survival *jointly*?

The common pipeline is **two-step**: run unsupervised factor analysis (PCA / EBMF), then test factors for survival *post hoc*.

- Unsupervised axes track the **largest expression variance** — often batch or housekeeping signal

- A program explaining little variance but strongly prognostic gets **missed**
- **Supervision** steers factors toward survival-relevant structure *during* estimation

A program explaining 15% of variance may be pure batch effect; one explaining 3% may cleanly separate survivors. Supervision prioritizes the second.

One background slide. The clinical motivation is unchanged. The methodological motivation — joint over two-step — is the thread of the whole talk, and I back it with both a synthetic comparison and a real-data external comparison against the unsupervised two-step (EBMF  $\rightarrow$  Cox) baseline.

## Methods

### The model — two coupled components

$$\underbrace{Y = LF^\top + E, \quad E_{ij} \sim N(0, \tau_j^{-1})}_{\text{genomics: low-rank+gene-specific noise}} \quad \underbrace{h_i(t) = h_0(t) \exp(\eta_i)}_{\text{Cox proportional hazards}}$$

| Quantity | Dim          | Meaning                                                |
|----------|--------------|--------------------------------------------------------|
| $Y$      | $n \times p$ | expression matrix                                      |
| $L$      | $n \times K$ | patient program-activation scores                      |
| $F$      | $p \times K$ | gene weights per program                               |
| $\beta$  | $K \times 1$ | survival (log-HR) coefficients                         |
| $\tau_j$ | scalar       | gene-specific noise precision                          |
| $\eta_i$ | scalar       | patient log-risk score ( $L_i\beta$ or $(Y_iF)\beta$ ) |

#### Shared loadings couple the two

The **same** latent structure drives low-rank reconstruction of  $Y$  **and** the survival risk score. A program is **prognostic only if its**  $\beta_k \neq 0$ ; otherwise it explains expression alone.

#### Non-parametric baseline hazard

$h_0(t)$  is left **unspecified** — it cancels in the Cox partial likelihood. No Weibull / exponential shape assumption; only proportional hazards.

Two equations, one shared L. Every symbol is defined in the table, including the risk score eta. The baseline hazard is non-parametric — a specific April question — so  $h_0(t)$  drops out of the partial likelihood and we make no parametric shape assumption.

## Two linear predictors: L vs (YF)

**Loadings predictor** —  $\eta = L\beta$

- Risk uses the **learned** patient scores  $L$  directly
- Survival shapes  $L$  during estimation (fully joint)
- Scoring a **new** patient requires **re-deriving** their  $L$  (an OLS projection) — an **approximation** of the trained scores

**Projection predictor** —  $\eta = (YF)\beta$

- Risk uses the **direct genomic projection**  $YF$
- **Exact** out-of-sample scoring:  $\eta_{\text{new}} = (Y_{\text{new}}F)\beta$  — the *identical* formula at train and test, no train/test score mismatch

**Priors** — **every block is explicit**.  $q(\cdot)$  denotes the **variational (approximate posterior)** for each block, fit by empirical Bayes at every update:

- $q(L_{\cdot k}), q(F_{\cdot k})$  — **point-exponential** (non-negative spike-and-slab):  $g(\theta) = \pi_0 \delta_0 + (1 - \pi_0) \text{Exp}(\lambda)$ ,  $\theta \geq 0$
- $q(\beta_k)$  — **normal** (Gaussian slab, no spike):  $g(\beta) = N(0, \sigma^2)$
- $q(\tau_j)$  — **closed-form MLE**: no EBNM, no prior tuning

Sparsity  $\pi_0$ , scale  $\lambda$ , and slab variance  $\sigma^2$  are all estimated from the data (**ebnm**) at each step.

💡 Why we report both

$(YF)\beta$  gives leakage-free **exact** external scoring  $\rightarrow$  the recommended model;  $L\beta$  is the fully-joint reference.

The key methods update since April: two linear predictors.  $L$  is the original fully-joint form but needs  $L$  re-derived for new patients.  $(YF)$  scores new patients with the identical formula — no mismatch. Priors are now correct and explicit: non-negative point-exponential on  $L$  and  $F$ , a Gaussian (normal) prior on  $\beta$ , and a closed-form MLE for  $\tau$ . Note these are NOT point-normal.

## The objective: ELBO & variational family

**Mean-field variational family** — each block factorizes independently:

$$q(L, F, \beta, \tau) = \prod_{k=1}^K q(L_{\cdot k}) \prod_{k=1}^K q(F_{\cdot k}) \prod_{k=1}^K q(\beta_k) \prod_{j=1}^p q(\tau_j)$$

The **ELBO** (evidence lower bound) is the objective CAVI maximizes — a lower bound on the log marginal likelihood. For the recommended **projection** model:

$$\text{ELBO} = \underbrace{E_q[\mathcal{L}_{\text{gen}}]}_{\text{expression fit}} + \underbrace{\lambda E_q[\mathcal{L}_{\text{Cox}}]}_{\text{survival fit}} - \underbrace{\sum_{\text{blocks}} D_{\text{KL}}(q \parallel \text{prior})}_{\text{prior regularization}}$$

- **Genomics fit**  $E_q[\mathcal{L}_{\text{gen}}] = \sum_j E_q \log p(Y_{\cdot j} \mid L, F_j, \tau_j)$  — Gaussian reconstruction of  $Y$  by  $LF^\top$

- **Survival fit**  $E_q[\mathcal{L}_{\text{Cox}}]$  — a **2nd-order Taylor** expansion of the Cox partial log-likelihood in  $\eta = (YF)\beta$ , making the survival term conjugate to the EBNM updates
- **KL term** penalizes each posterior  $q$  away from its prior — the adaptive shrinkage
- The **shared**  $L$  enters both likelihood terms — that coupling *is* the supervision
- The ELBO **balances** expression and survival automatically — **no tuning parameter**  $\alpha$  ( $\lambda = 1$  in the recommended fits)

**What the ELBO is for:** it is the **fitting objective** — CAVI maximizes it, and a fit is declared converged when it stops improving. **It is *not* used to choose the number of programs** — that is done by cross-validated C-index (next: *Selecting K*). ::: { .notes } The variational family is mean-field and factor-wise, written out at the top — a direct April request. The ELBO has three pieces: Gaussian genomics reconstruction, the Cox survival term via a second-order Taylor expansion that keeps the updates conjugate, and a per-block KL penalty that is the adaptive shrinkage. I show the survival term for the projection (YF) model, which is the recommended one. The shared L in both likelihood terms is the supervision, and the ELBO balances the two with no to tune. :::

## Inference: factor-wise CAVI

**Coordinate ascent (Gauss–Seidel)** — per outer iteration, for each factor  $k$ :

$$\beta_k \rightarrow L_{\cdot k} \rightarrow F_{\cdot k}, \quad \text{then } \tau \text{ once per sweep}$$

- Each update is an **EBNM** sub-problem: pseudo-data  $x = B/A$ , scale  $s = 1/\sqrt{A}$ , then **ebnm** returns the posterior mean **and** variance
- $L$  has a **dual-source precision** (genomics + survival):

$$A_{L,i} = \underbrace{\sum_j \tau_j E[F_{jk}^2]}_{\text{expression}} + \underbrace{W_{ii} E[\beta_k^2]}_{\text{survival}}$$

where  $W_{ii}$  is the patient's Cox working weight

- The empirical-Bayes prior  $\hat{g}$  is **re-estimated every update**  $\rightarrow$  adaptive shrinkage per factor / per gene; deflate the residual after each factor

**Convergence — in plain English.** After a 5-iteration burn-in, stop when the **ELBO stops improving** — its relative change between iterations falls below  $10^{-5}$ :

$$\frac{|\text{ELBO}^{(t)} - \text{ELBO}^{(t-1)}|}{|\text{ELBO}^{(t-1)}|} < 10^{-5}$$

- The ELBO is the **single objective** being maximized, so this directly detects a stationary point of the variational optimization
- Mean absolute changes in  $L$  and  $\beta$  are tracked alongside as **diagnostics** (both settle below  $10^{-3}$  at convergence)

### ! Risk-direction sign correction

Cox concordance is invariant to a global sign flip, so an unconstrained fit can land at  $C < 0.5$ . **Fix:** orient each program against the **training** concordance, then apply the *same* orientation at test. (Used to define program activation in the survival results.)

CAVI updates one factor at a time in Gauss-Seidel order; each block reduces to an EBNM call with pseudo-data  $x=B/A$  and scale  $s=1/\sqrt{A}$ . The L precision is dual-source — a genomics piece summing tau times  $E[F^2]$ , plus a survival piece  $W_{ii}$  times  $E[\beta^2]$  — so survival genuinely shapes the loadings. Convergence requires both L and beta to settle below  $1e-3$  in MEAN absolute change; averaging avoids a single oscillating factor blocking the stop. The sign correction orients each program by training concordance and reuses it at test.

## Preprocessing is crucial: per-platform normalization

**The  $p \gg n$  scaling problem.** Merging platforms (RNA-seq, microarray, proteomics) leaves each gene on a different scale. The survival precision for program  $k$  is

$$A_k = \sum_i w_i (YF)_{ik}^2,$$

where  $w_i$  = the patient's **Cox working weight** (curvature from the survival Taylor step) and  $(YF)_{ik}$  = patient  $i$ 's **projection score** on program  $k$ . Without correction, between-platform variance inflates  $(YF)$  unevenly, **swamps**  $A_k$ , and the survival coefficients **collapse to**  $\beta \rightarrow 0$ .

**The fix — normalize *per platform, before merging*:**

1. Log2 transform (RNA-seq only)
2. **Per-cohort gene selection** (combined rank of mean  $\times$  variance), top-3000 per cohort, then **intersect**  $\rightarrow$  **2064 genes** in the merged training matrix
3. **Per-platform z-standardization** (each cohort centered/scaled on its own)
4. Row-bind into the merged training matrix

❗ Skip per-platform normalization  $\rightarrow$  collapses

In an 18-configuration benchmark, **10 of 12** pipelines that do **not** z-standardize per platform before merging drive  $\beta \rightarrow 0$  — for *both* linear predictors.

💡 No leakage

Normalization is fit **within each training cohort, before** any split; held-out cohorts get the *same* per-cohort recipe at scoring time.

The single most important practical finding since April.  $A_k$  is the survival precision for program  $k$  — a sum over patients of the Cox working weight  $w_i$  times the squared projection score. Define both pieces. Without per-platform normalization the projection scales blow up across platforms,  $A_k$  is dominated by between-platform variance, and beta collapses. The fix is per-platform z-standardization before merging, which leaves 2064 genes. 10 of 12 alternatives collapse. Normalization is fit per training cohort before any split — no leakage.

## Selecting the number of programs: $K$ and $K_{\text{eff}}$

Two distinct filters — the deck keeps them separate:

### 1 · Retained programs $K$ — by cross-validation

- **5-fold CV** on the training data; record the held-out **C-index** at each  $K$  in a grid
- **1-SE rule:** pick the smallest  $K$  whose mean CV C-index is within one standard error of the best
- **Biological floor**  $K \geq 3$ : a *programs* comparison needs at least three (aligned with the lab's penalized survival-MF work)
- $\Rightarrow$  recommended preprocessing selects  **$K = 7$**

## 2 · Survival-active programs $K_{\text{eff}}$ — by $|\hat{\beta}|$

- Fit the model at the retained  $K = 7$
- A program is **survival-active** if its posterior coefficient clears the activity threshold:  $|\hat{\beta}_k| > 10^{-3}$
- The rest have  $\hat{\beta}_k$  shrunk to  $\approx 0$  — they **explain expression only**
- $\Rightarrow K_{\text{eff}} = 2$  of the 7 retained programs carry survival signal

💡 Read both numbers together

$K$  is *how many programs the model keeps* (reconstruction; chosen by CV).  $K_{\text{eff}}$  is *how many of those are prognostic* (chosen by  $|\hat{\beta}|$ ). For the recommended model:  
 $K = 7 \rightarrow K_{\text{eff}} = 2$ .

This slide answers two questions the audience will have. First, how do we choose  $K$  — by 5-fold cross-validated C-index with the 1-SE rule and a biological floor of 3, giving  $K=7$ . Second, what makes a program “active” — its survival coefficient must exceed  $1e-3$ ; only 2 of the 7 do, so  $K_{\text{eff}}=2$ .  $K$  is reconstruction capacity;  $K_{\text{eff}}$  is the prognostic subset. Always quote both.

### L extension: cohort-membership indicator

**Idea.** Append corner-point dummy columns (one per non-reference cohort) to  $L$ , with their survival coefficient **fixed at**  $\beta_{\text{cohort}} \equiv 0$ , so survival risk is driven only by the **biological** programs while the dummies absorb platform offsets in the genomics term.

- Updated via a Normal-conjugate step (cohort-row  $F$  update)
- The forced dummy consumes **one degree of freedom**, so we select **one additional factor** ( $K = 7 \rightarrow 8$ ) to keep the same number of biological programs

**When it helps — and when it doesn't**

- At low  $K$  with weak normalization, the indicator can rescue platform-confounded fits
- Under **per-platform z-standardization**, the offset is *already* removed — the indicator adds a factor slot for **no benefit**

### Recommendation

With per-platform z-standardization, adding the cohort indicator **moves mean external C-index down** from **0.636** to **0.614** (and  $K : 7 \rightarrow 8$ ).

**We therefore recommend no cohort indicator** — normalization already absorbs the platform offset.

The cohort indicator appends dummy columns to  $L$  with beta fixed at zero, so survival uses only biological programs and the dummies soak up platform offsets. Because the forced dummy costs a degree of freedom, we add one more factor. It helps when normalization is weak, but under per-platform z-std the offset is already gone — it just costs a slot and slightly lowers external C, from 0.636 to 0.614. So the recommendation is to leave it off.

# Results

## Data — Synthetic Validation & PDAC Cohorts

### Multi-cohort simulation

**EBMF-grounded ground truth:** shared factors (present in all studies) + study-specific factors; **shared factors carry the survival signal**, study-specific do not.

### Scenarios

- *All Shared Factors* — every program is shared & prognostic
- *All Cohort-Specific Factors* — negative control; no shared signal ( $\eta \equiv 0$ )
- *Shared & Cohort-Specific Factors* — the realistic mix

### Model Arms

- EBMF (unsupervised two-step baseline)
- $\text{YFB\_base} \cdot \text{YFB\_cohort} \cdot \text{LB\_base} \cdot \text{LB\_cohort}$

### PDAC cohorts — External Validation

- **Train:** TCGA-PAAD + CPTAC ( $n \approx 273$ , RNA-seq + proteomics)
- **Validate:** 5 fully held-out cohorts, scored by the **exact** projection formula  $\eta = (Y_{\text{new}}F)\beta$

| Held-out cohort | Platform   |
|-----------------|------------|
| Dijk            | RNA-seq    |
| Moffitt         | Microarray |
| PACA-AU (array) | Microarray |
| PACA-AU (seq)   | RNA-seq    |
| Puleo           | Microarray |

Two datasets. The simulation is a completed validation study with full results — three scenarios (all-shared, the all-cohort-specific negative control, and the realistic shared-plus-specific mix), five model arms including the unsupervised EBMF baseline, five seeds each. On real data we train on TCGA + CPTAC and validate on five fully held-out cohorts spanning microarray and RNA-seq, scored by the exact projection formula.

Synthetic Results — Shared & Cohort-Specific Factor Recovery

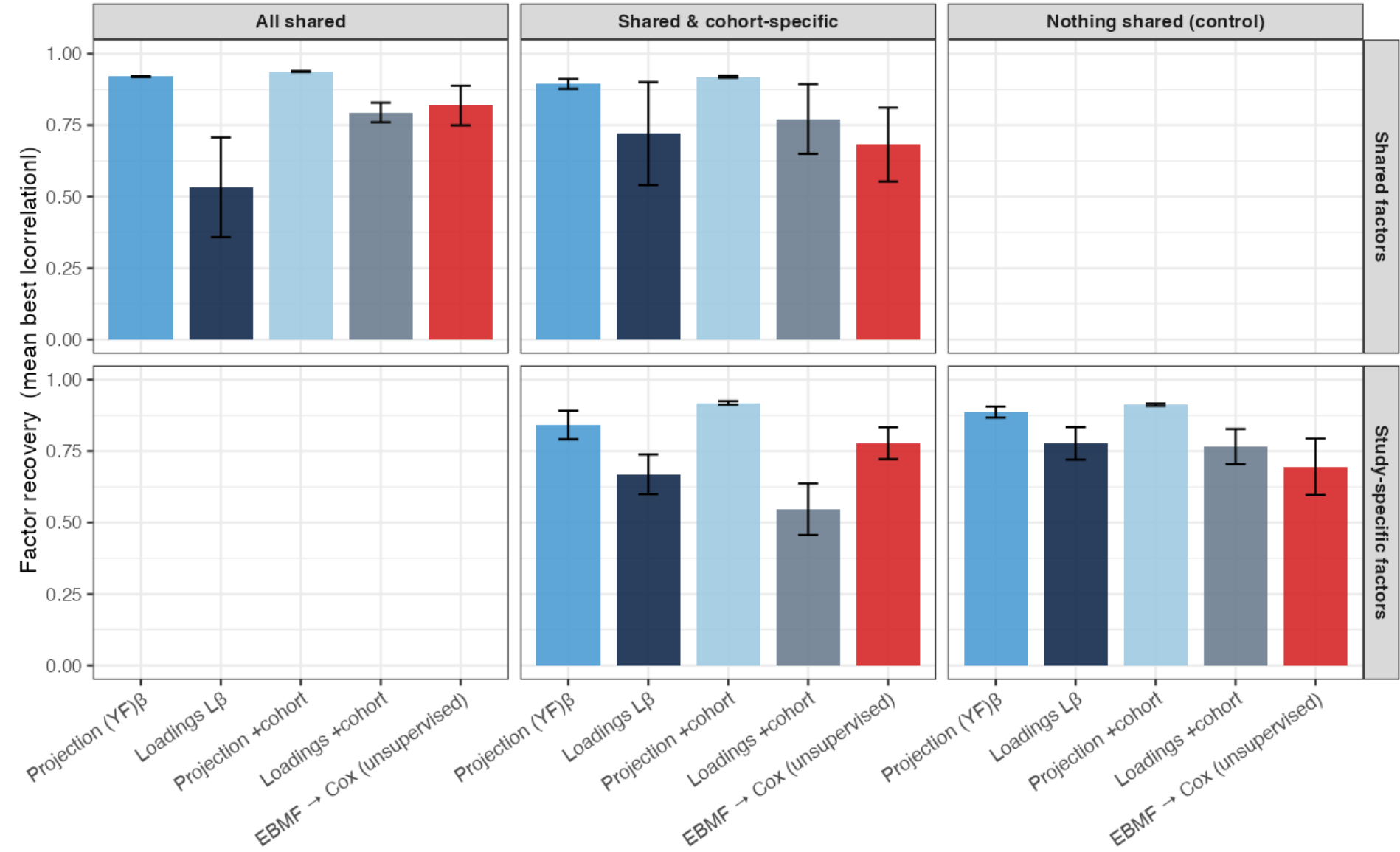


Figure 1: **Figure 1.** Factor recovery = mean best |correlation| between each estimated gene program and the true program it matches, by model arm and scenario (mean  $\pm$  SE over 5 seeds). Top row: shared (prognostic) factors; bottom row: study-specific factors.



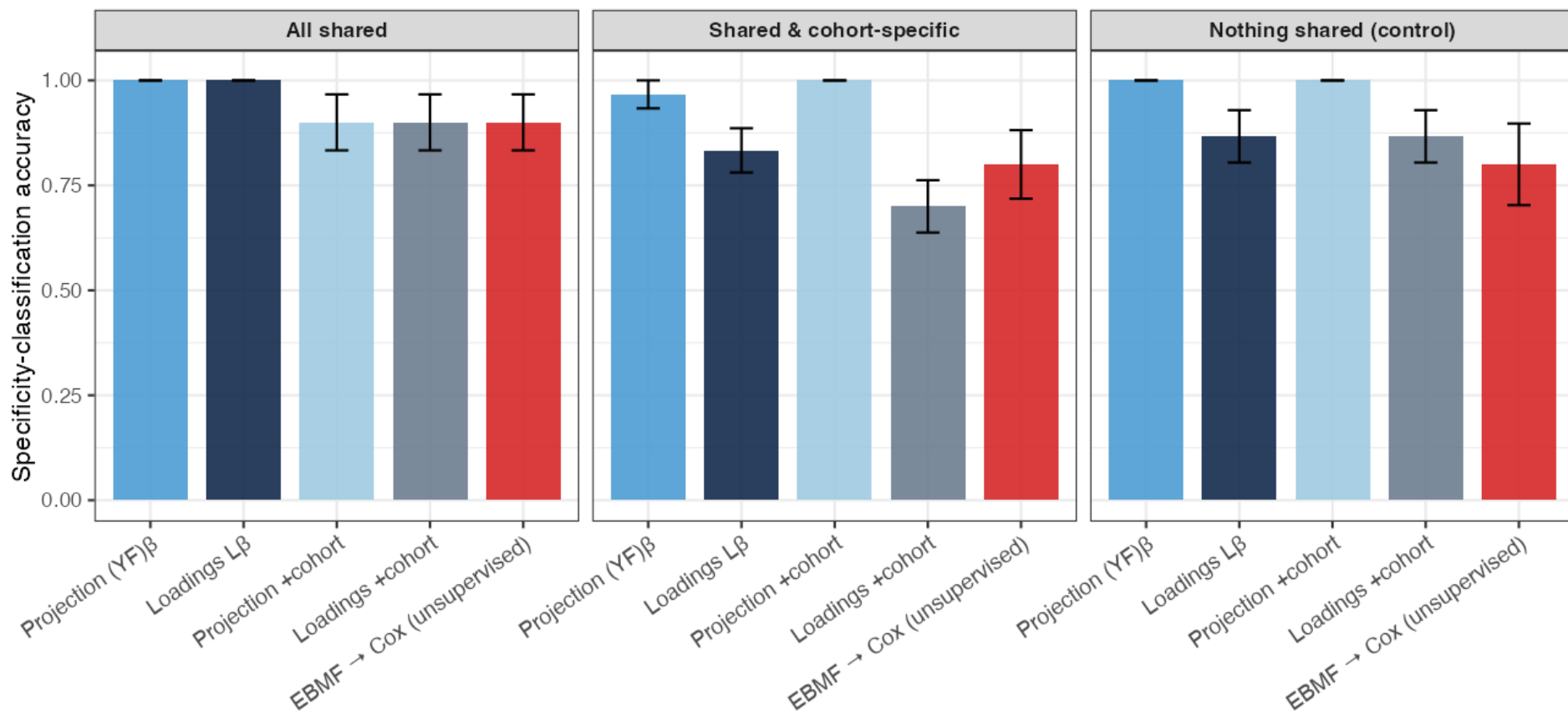


Figure 2: **Figure 2.** Specificity-classification accuracy = fraction of recovered factors correctly *typed* as shared vs. study-specific, from per-cohort loading energy alone (mean  $\pm$  SE, 5 seeds).

- **Two complementary questions.** Figure 1 asks *how faithfully* each gene program is reconstructed (a correlation); Figure 2 asks whether the model *correctly labels* each program as shared or study-specific (a classification). High recovery with low typing accuracy would mean good programs, wrong structure — so we report both.
- The **projection model** recovers shared factors at  $|r| \approx 0.90$ – $0.92$  and types them with accuracy up to **1.00** — it natively separates shared from study-specific structure, which is *why* the cohort indicator adds little.

Two metrics, two questions. Recovery (Figure 1) is a correlation: how well is each program reconstructed. Specificity accuracy (Figure 2) is a classification: does the model correctly call each factor shared vs study-specific, using only per-cohort loading energy. We need both because a model could reconstruct programs well yet mislabel their structure. The projection model scores high on both, which is the reason the explicit cohort indicator buys little.

## Synthetic Results — Factor Recovery Heatmap

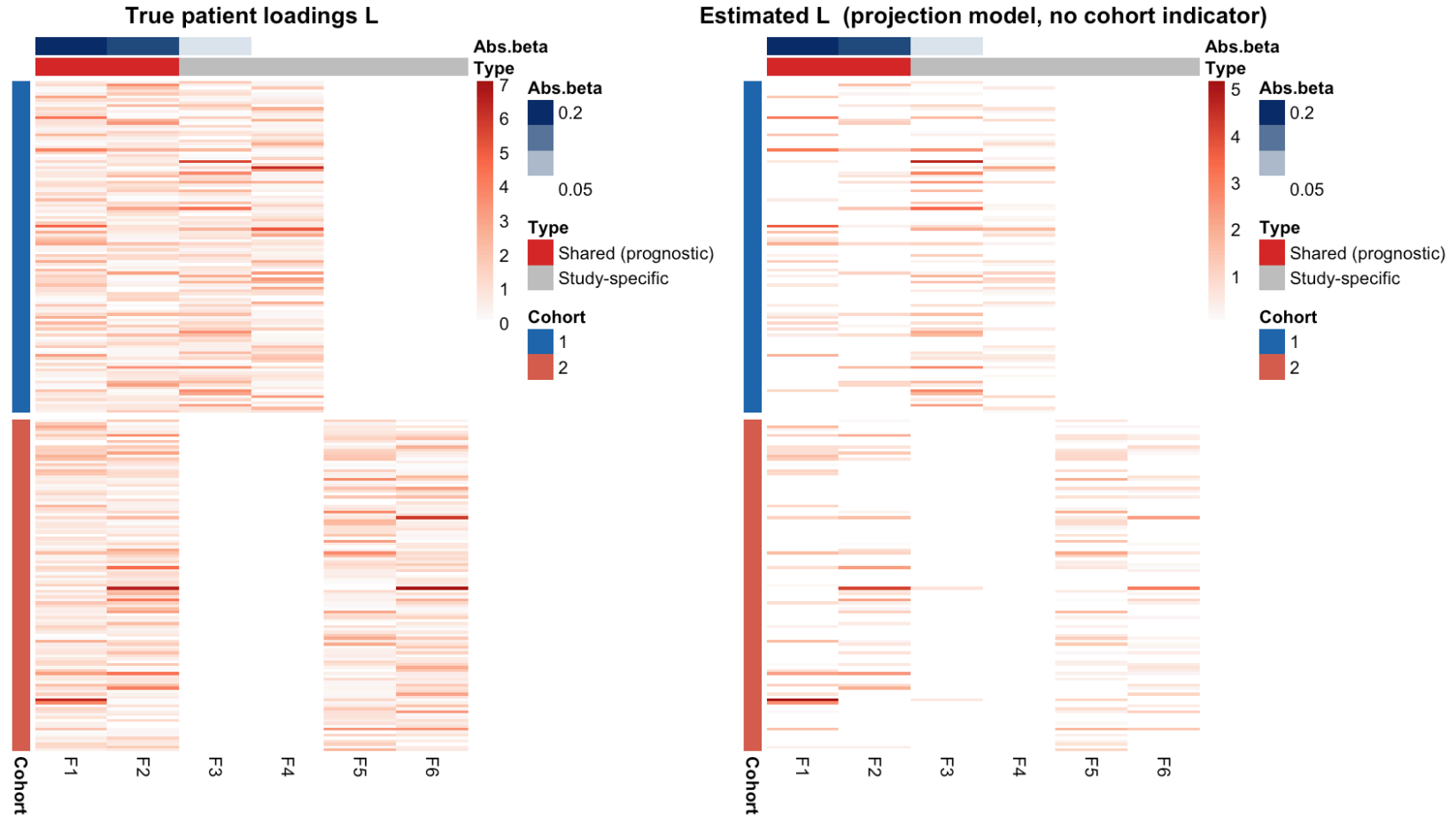


Figure 3: **Figure 3.** Patient-loading matrix  $L$  in the realistic *Shared & Cohort-Specific* scenario (hybrid, seed 1). Rows = patients grouped by cohort (left bar); columns = factors. Top annotations: true factor type (shared/prognostic vs. study-specific) and  $|\hat{\beta}|$ . **Left:** true block-sparse  $L$ . **Right:** estimated  $L$  from the projection model — recovering shared factors (active in *both* cohorts) and study-specific factors (active in one), with **no cohort indicator**.

- Shared factors light up across **both** cohort blocks; study-specific factors light up in **one** — the model recovers the intended structure directly from the data.

This heatmap makes the recovery concrete. On the left is the true loading matrix with its block-sparse structure: shared factors active in both cohort blocks, study-specific factors active in only one. On the right, the estimated loadings from the projection model reproduce that structure with no indicator columns — the visual basis for the recovery and specificity numbers on the previous slide.

## Synthetic Results — SSBMF outperforms EBMF+Cox

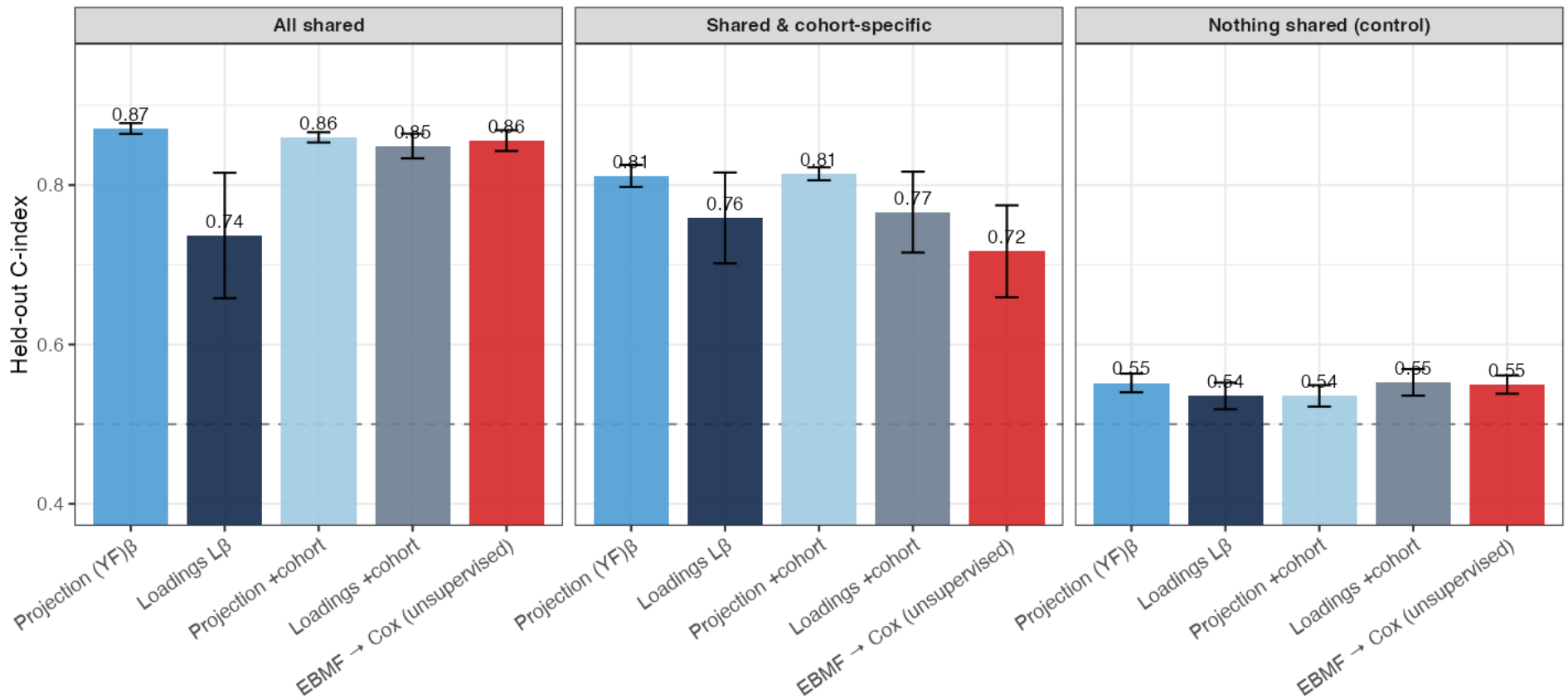


Figure 4: **Figure 4.** Held-out C-index by model arm within each scenario (mean  $\pm$  SE over 5 seeds; dashed line = chance). The unsupervised EBMF  $\rightarrow$  Cox baseline is shown alongside the supervised arms.

**Hybrid (Shared & Cohort-Specific), mean C-index (over 5 seeds):**

- **Projection (YF) $\beta$ : 0.81**
- **Loadings  $L\beta$ : 0.76**
- **Unsupervised EBMF  $\rightarrow$  Cox: 0.72**

**Negative control (All Cohort-Specific):** all arms at  $\approx 0.54$  — no false prognostic signal manufactured when none exists.

### Takeaway

Joint supervision beats the unsupervised two-step approach on data with known ground truth — and invents no signal when there is none.

The headline of the synthetic study. In the realistic hybrid scenario the supervised projection model reaches 0.81, ahead of the loadings model at 0.76 and the unsupervised EBMF-then-Cox two-step at 0.72 — averaged over five seeds. Critically, in the all-cohort-specific control every arm sits at chance, so the method does not fabricate prognostic signal. That is the clean demonstration that supervision helps.

## PDAC Cohorts — Prognostic Gene Programs

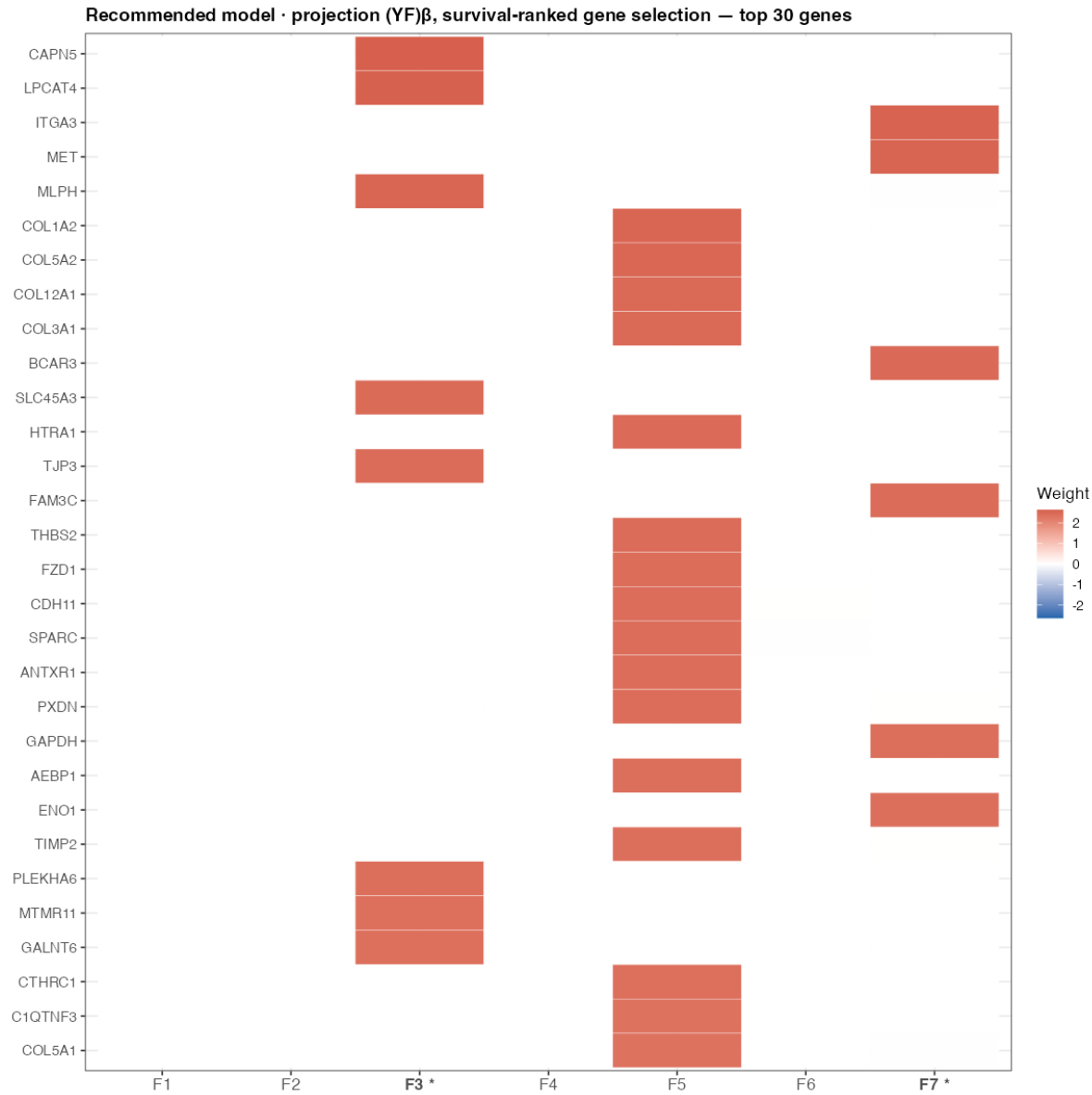


Figure 5: **Figure 5.** Gene-weight heatmap (top 30 genes) for the **recommended model**: projection  $(YF)\beta$ , per-platform z-standardization, survival-ranked gene selection (2064 genes), **no** cohort indicator,  $K = 7$ . Weights are **non-negative** (point-exponential prior). Survival-active programs ( $|\hat{\beta}| > 10^{-3}$ ) are starred.

- $K = 7$  programs **retained** (CV + 1-SE, floor  $K \geq 3$ );  $K_{\text{eff}} = 2$  **survival-active** — Program 3 (**protective**) and Program 7 (**adverse**)

- The *largest* expression program (collagen/ECM/stromal) is **not** survival-active — supervision separates expression-only structure from prognostic structure

On real data, the recommended model keeps seven programs by CV, but only two are survival-active: Program 3 (protective) and Program 7 (adverse). Always quote K=7 retained, K\_eff=2 active. Note the largest expression program — the stromal collagen block — is NOT survival-active, exactly the separation supervision is supposed to give you. Weights are non-negative because L and F use a point-exponential prior.

## PDAC Cohorts — Prognostic Association

The two survival-active programs stratify patients (split at median **projection score**  $(YF)_k$  — the quantity the model scores on):

- **Adverse — Program 7** (higher activation → shorter survival), log-rank  $p = 0.00047$ 
  - top genes: *ITGA3*, *MET*, *BCAR3*, *FAM3C*, *GAPDH*, *ENO1*, *TPI1*, *PGK1*
- **Protective — Program 3** (higher activation → longer survival), log-rank  $p = 0.0077$ 
  - top genes: *CAPN5*, *LPCAT4*, *MLPH*, *SLC45A3*, *TJP3*, *PLEKHA6*, *MTMR11*, *GALNT6*

Proportional-hazards assumption not violated (Schoenfeld test).

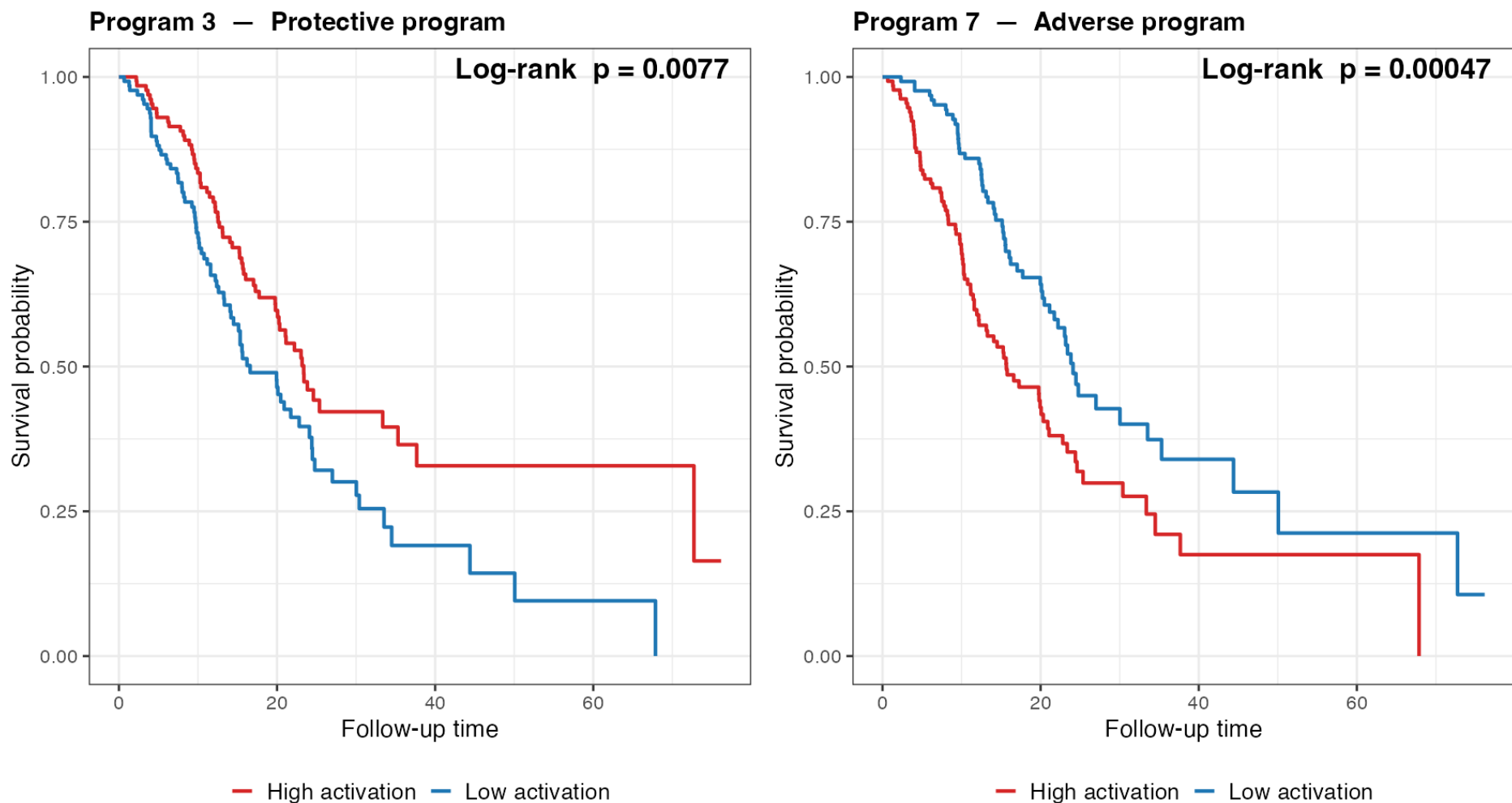


Figure 6: **Figure 6.** Kaplan–Meier curves for the two survival-active programs, patients split at the median projection score  $(YF)_k$  in the training cohorts. The adverse program’s high-activation group has worse survival; the protective program’s, better. Direction read empirically from the curves.

The two active programs stratify patients into clear survival groups. We split by the projection score  $(YF)_k$  — the quantity the risk model uses, not the raw loading. Program 7, the adverse program, carries MET, ITGA3, and glycolytic genes — biologically a classic aggressive signature — and its high-activation group does worse,  $p=0.0005$ . Program 3, protective, carries epithelial differentiation markers,  $p=0.008$ . Directions are read off the curves. Schoenfeld test shows PH holds. If asked: the joint-model beta sign for program 7 is negative even though its marginal direction is adverse — a suppression effect among correlated programs; we label by the empirical survival direction the curve shows.

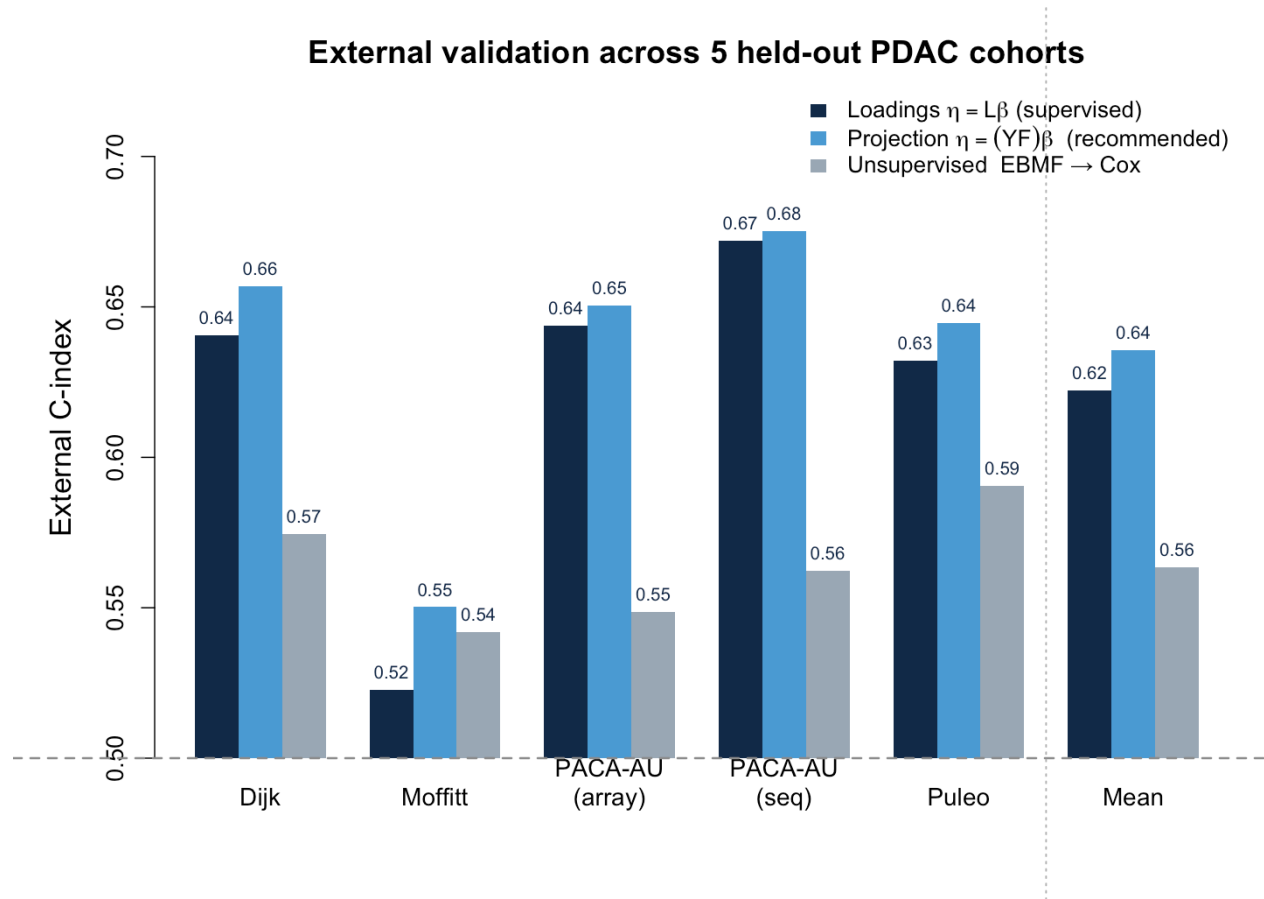


Figure 7: **Figure 7.** External C-index across the five held-out PDAC cohorts (+ mean) for three arms: loadings  $L\beta$ , projection  $(YF)\beta$ , and the unsupervised EBMF  $\rightarrow$  Cox two-step. Dashed line = chance.

- **Projection (YFB) model mean C = 0.636** · loadings (LB) model 0.622 · EBMF  $\rightarrow$  Cox 0.564, across the five held-out cohorts
- The projection-based supervised joint model **outperforms the unsupervised two-step approach across all 5 external PDAC cohorts**, with  $K = 7$  (CV + 1-SE, floor  $K \geq 3$ )  $\rightarrow K_{\text{eff}} = 2$

The real-data analog of the synthetic slide. Same three arms, now scored on five fully held-out cohorts. The projection model averages 0.636 and beats the unsupervised EBMF-to-Cox two-step in all five cohorts. I also lean on parsimony and interpretability — two survival-active programs and exact external scoring, versus roughly 20 EBMF factors.



Model Comparison Summary

|                          |                                                                          |         |
|--------------------------|--------------------------------------------------------------------------|---------|
| Model                    | Linear preprocessing predictor                                           | C-index |
| Projection (recommended) | (YF)-platform z-std + survival-ranked gene selection                     | n       |
| Loadings                 | Lper-platform z-std + survival-ranked gene selection                     | n       |
| Projection + cohort ind. | (YF)-platform z-std + survival-ranked gene selection                     | ye      |
| Unsupervised two-step    | EBM-platform z-std + survival-ranked gene selection<br>—selection<br>Cox | n       |

Synthetic C = mean held-out C-index, hybrid scenario (5 seeds). Real ext. C = mean across 5 held-out PDAC cohorts. **K** = CV + 1-SE selected (biological floor K 3); **K\_eff** = survival-active programs ( $\uparrow$  > 1e-3). Gene selection ranks genes by combined mean  $\times$  variance per cohort.

- **Recommendation:** projection predictor (YF) $\beta$   $\cdot$  per-platform z-standardization  $\cdot$  survival-ranked gene selection  $\cdot$  **no** cohort indicator
- Wins on both axes — best synthetic **and** real external C-index — with the most **parsimonious, interpretable** survival structure ( $K = 7 \rightarrow K_{\text{eff}} = 2$ ) and **exact** external scoring

**i** On gene selection

The combined-rank selection recipe was **motivated by** the lab’s penalized survival-MF (DeSurv) preprocessing; here it is applied within the fully Bayesian model.

The summary table describes every model by its characteristics — no internal code names. Projection predictor with per-platform z-std and survival-ranked gene selection, no cohort indicator, is the recommendation: best on both synthetic and real, most parsimonious with only two survival-active programs, and exact external scoring. The gene-selection recipe was motivated by the lab’s penalized DeSurv preprocessing but applied inside the Bayesian model.

## Impact / key findings

1. **Joint supervision beats the unsupervised two-step** — by  $\sim 0.09$  C-index on synthetic data with known truth (0.81 vs 0.72, hybrid), and across **all five** held-out PDAC cohorts on real data.
2. **Preprocessing is decisive** — per-platform z-standardization before merging is the difference between a working model and  $\beta \rightarrow 0$ ; **10 of 12** alternatives collapse.
3. **Parsimony with signal** — the recommended model carries just  $K_{\text{eff}} = 2$  **survival-active programs** (vs  $\sim 20$  unsupervised factors) yet matches the field, with **exact, leakage-free external scoring**.
4. **Mean external C-index = 0.636** across five independent PDAC cohorts — fully Bayesian throughout (adaptive priors, no  $\alpha$  tuning, uncertainty-aware).

## Limitations

- **Small external cohorts give noisy estimates** — per-cohort C ranges down to 0.55 (Moffitt), near chance
- **Real-data margin over unsupervised EBMF  $\rightarrow$  Cox is modest** (EBMF is already 0.63 within-training); the *synthetic* gap (0.81 vs 0.72) is the cleaner demonstration of supervision's value
- The two programs are **statistically prognostic** — **biological annotation (pathway enrichment) is the next step**, not yet done

Four takeaways as plain numbered lines, then an honest limitations beat. Small external cohorts (Moffitt near chance) give noisy C-index; the real-data margin over EBMF is modest because EBMF is already strong within-training, so I lean on the cleaner synthetic gap and on parsimony/exact-scoring; and the programs are statistically prognostic but not yet biologically annotated — that's the pathway enrichment next step. Supervision helps — clearly on synthetic, across all five cohorts on real. Preprocessing is decisive. Parsimony with signal: two active programs, exact external scoring, performance competitive with the field, all fully Bayesian with adaptive priors and no alpha to tune.

## Future directions & summary

### Addressed since April

- **CV-based K selection** — 5-fold CV, 1-SE rule
- **Multi-cohort shared-factor validation** — completed simulation study

### Next steps

1. **Pathway enrichment** on the two survival-active programs — assign biology to the statistical factors
2. **Manuscript preparation** — methods-primary framing; PDAC as the application

## Summary

1. Joint supervised factorization, two linear predictors; **point-exponential priors on  $L, F$ , a normal prior on  $\beta$** , non-parametric baseline hazard
2. **Preprocessing is crucial** — per-platform normalization prevents the survival precision from being dominated and  $\beta$  from collapsing
3. Validated on a completed multi-cohort simulation (hybrid C = 0.81 vs 0.72) **and** five external PDAC cohorts (mean C = 0.636)
4. **Recommended model**
  - Linear predictor: projection  $(YF)\beta$
  - L cohort indicator: No
  - Pre-processing: per-platform z-std
  - Selected  $K = 7$  ( $K_{\text{eff}} = 2$ ), mean external C = **0.636**

We addressed two April next-steps directly: CV-based K selection and the multi-cohort shared-factor study are both complete. Next: pathway enrichment to name the two active programs biologically, then the manuscript. Summary in four points — the joint model with the correct priors, the preprocessing fix stated as crucial for avoiding precision domination and beta collapse, validation on simulation plus five external cohorts with the numbers, and the recommended configuration broken out line by line.

## Discussion

### Thank You

Questions, suggestions, and feedback welcome!

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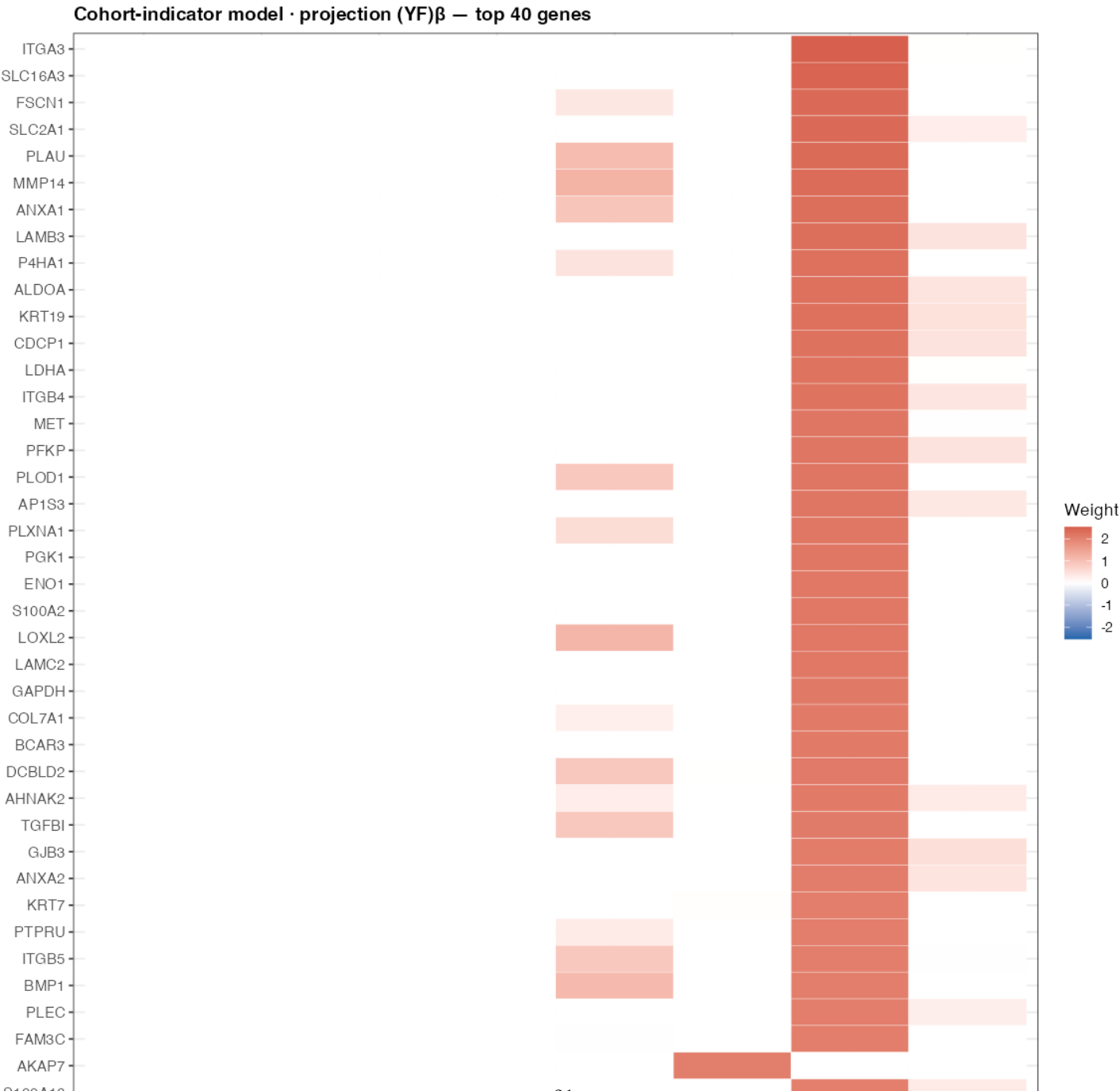
Department of Biostatistics, UNC Gillings School of Global Public Health

Open the floor. I have appendix slides on the cohort-indicator contrast and on convergence and the survival-activity magnitudes.



Appendix

Appendix — cohort-indicator contrast



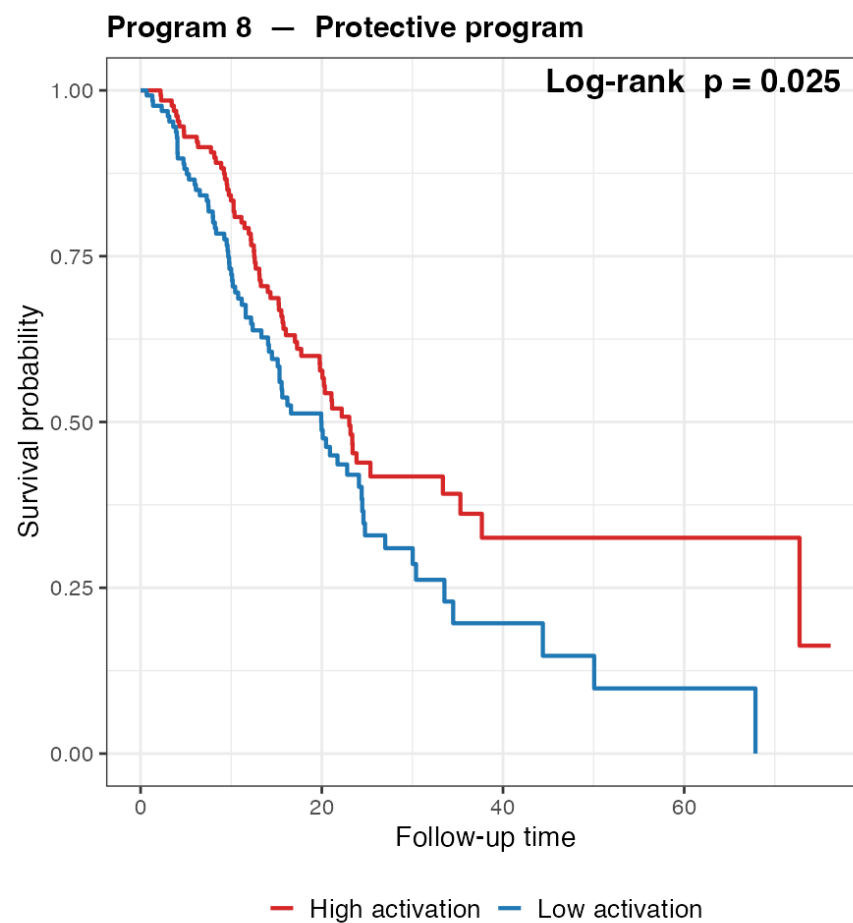
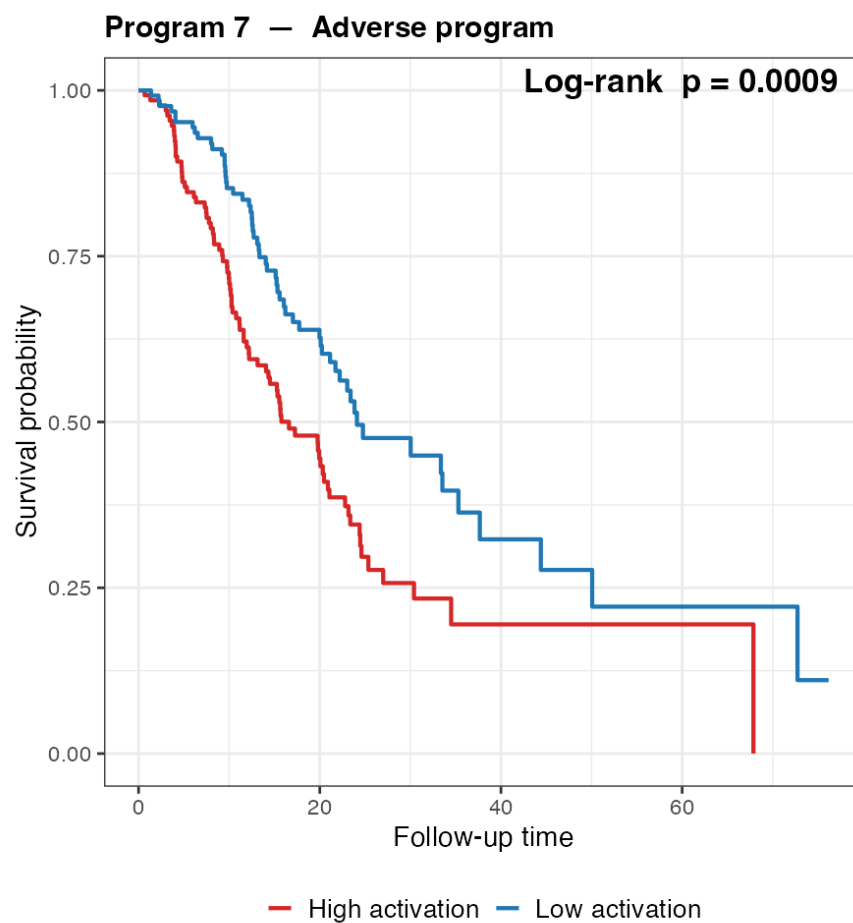


Figure 9: **Figure A2.** Survival stratification with the cohort-membership indicator, by projection score. The survival-active structure is essentially unchanged from the recommended model.

- Adding the indicator under per-platform z-std spends a factor slot ( $K : 7 \rightarrow 8$ ) **without** improving external C-index ( $0.636 \rightarrow 0.614$ )
- The biological survival-active structure is essentially unchanged — confirming the platform offset is already removed by normalization

## Appendix — convergence & survival-activity magnitudes

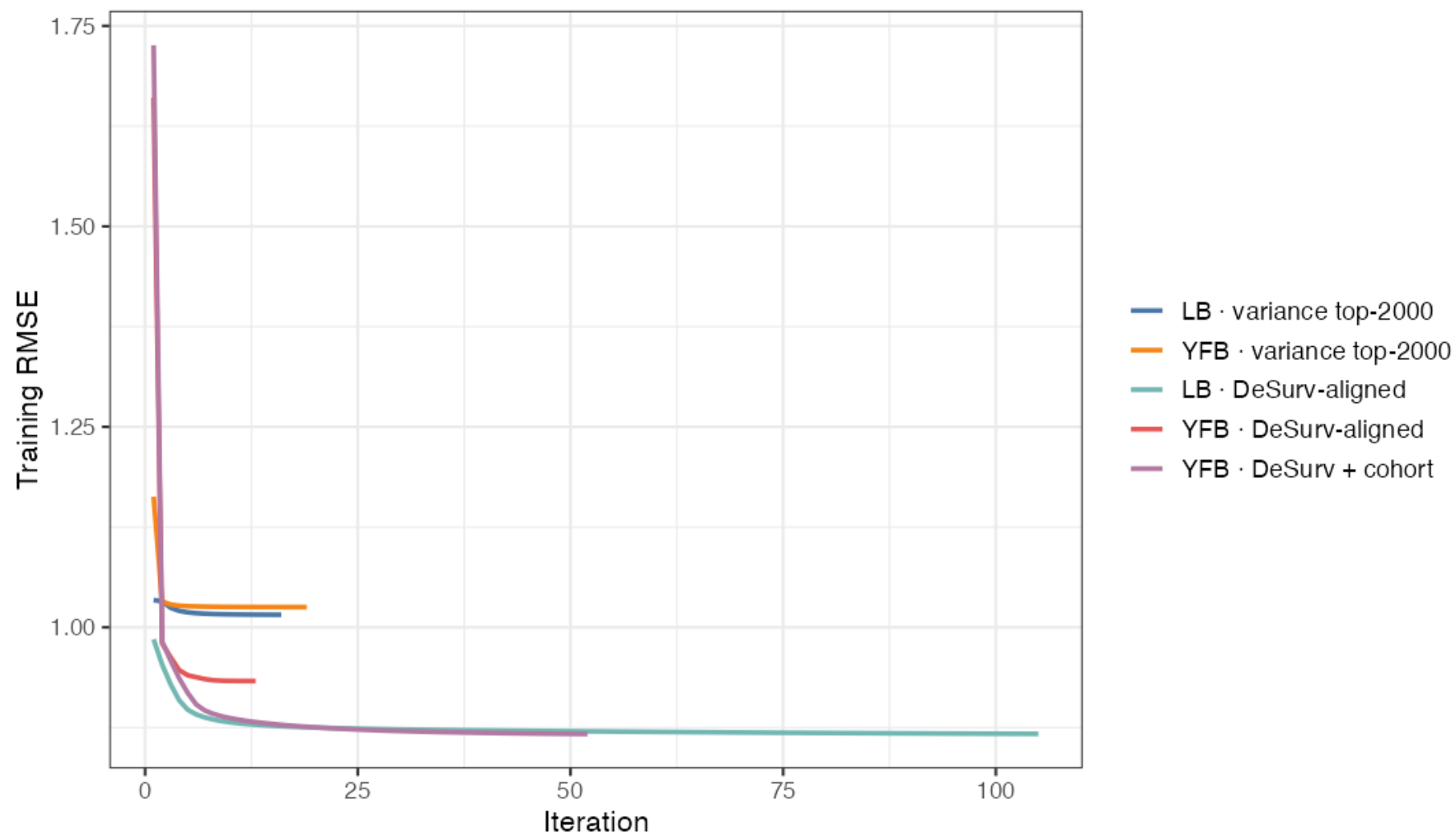


Figure 10: **Figure A3.** Training RMSE convergence traces across model configurations — RMSE stabilizes to a plateau; stopping is governed by the relative ELBO change falling below  $10^{-5}$  ( $\Delta L, \Delta \beta$  tracked as diagnostics).

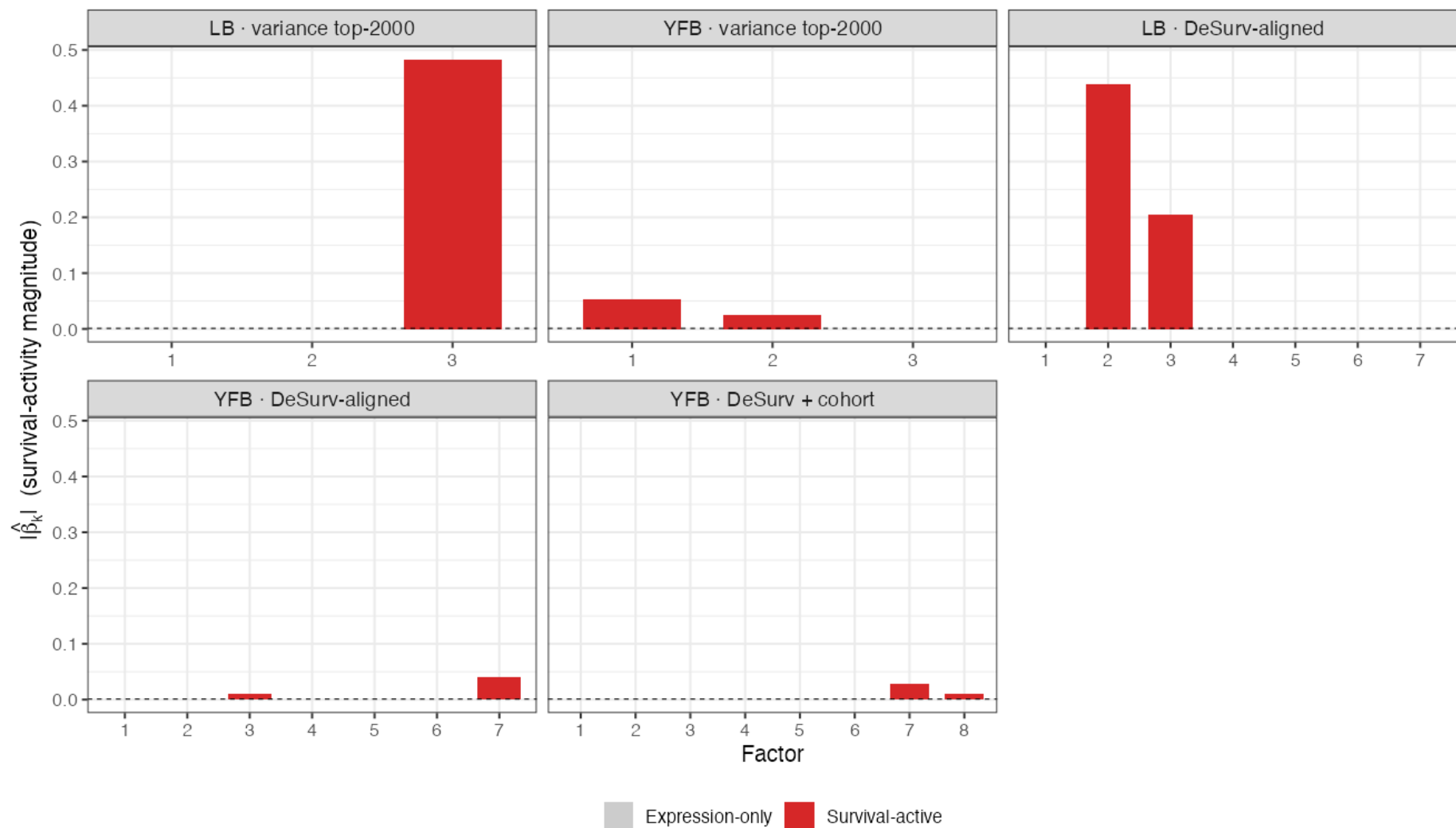


Figure 11: **Figure A4.** Survival-activity magnitude  $|\hat{\beta}_k|$  per program. Most programs fall below the activity threshold (expression-only); a small number clear it (survival-active).

- The spike-and-slab / shrinkage behavior is visible: most programs are shrunk below the survival-activity threshold, with only the survival-active ones retained

## Appendix — relation to DeSurv (Young et al., PNAS 2026)

**Same design & same data — SBMF is the Bayesian counterpart.** Both jointly fit supervised NMF + Cox with **non-negative gene programs**, share the same training set (TCGA + CPTAC,  $n = 273$ ) and the **same five external PDAC cohorts** (Dijk, Moffitt, PACA-AU array/seq, Puleo;  $n = 616$ ), score new



patients by **projection** (no retraining), select via **CV C-index + 1-SE**, and use the **combined-rank top-3000-per-cohort** gene selection. Both also find that the **highest-variance program is *not* the most prognostic**.

DeSurv objective:  $(1 - \alpha) \mathcal{L}_{\text{NMF}} - \alpha \mathcal{L}_{\text{Cox}}$ , with the survival gradient on the gene programs  $W$ .

**What SBMF changes**

|             | DeSurv                 | SBMF                                            |
|-------------|------------------------|-------------------------------------------------|
| Estimation  | penalized optimization | Bayesian variational (CAVI)                     |
| Tuned by BO | $k, \alpha, \lambda$   | <b>only</b> $K$ ( $\alpha, \lambda$ not tuned)  |
| Shrinkage   | fixed elastic-net      | empirical-Bayes priors (learned)                |
| Uncertainty | point estimates        | posterior mean + variance                       |
| Noise       | least-squares          | per-gene precision $\tau$ ; $E[L^2]$ correction |

- **DeSurv:**  $k = 3, \alpha = 0.334$ ; pooled external **HR/SD** = 1.50 (95% CI 1.31–1.72)
- **SBMF (YF) :**  $K = 7 \rightarrow K_{\text{eff}} = 2$ ; mean external **C \$=0.636**

**Honest caveat:** the two report different primary metrics (DeSurv: pooled HR/SD; SBMF: mean external C-index) and use slightly different normalization (DeSurv within-subject rank  $\rightarrow$  1970 genes; SBMF per-platform z-std  $\rightarrow$  2064). A **matched same-protocol head-to-head is a planned next step** — DeSurv is *not* re-run here.

This is the backup for the inevitable “how does this compare to DeSurv?” question. Be collegial and precise: SBMF is the Bayesian sibling of DeSurv — same supervised NMF+Cox idea, same training data and the same five external cohorts, same projection-based scoring and CV+1-SE selection, and SBMF borrowed DeSurv’s combined-rank gene selection. The genuine differences: DeSurv tunes k, alpha, and lambda by Bayesian optimization; SBMF only selects K and lets the empirical-Bayes priors learn the shrinkage, with no alpha or lambda to tune, plus posterior uncertainty and a heteroscedastic noise model. On numbers, DeSurv reports pooled HR per SD of 1.50; SBMF reports mean external C of 0.636 — different metrics, so a matched head-to-head on one protocol is the honest next step, not a claim of superiority today.